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In [1]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: conn = sqlite3.connect("module6_lab3.db")
```

```
In [3]: conn.execute("DROP TABLE IF EXISTS custome

conn.execute("""
CREATE TABLE customers (
    id INTEGER PRIMARY KEY,
    name TEXT,
    city TEXT,
    age INTEGER
)
""")
```

```
Out[3]: <sqlite3.Cursor at 0x7fbc7662e240>
```

```
In [ ]: conn.executemany("""
INSERT INTO customers (id, name, city, age
VALUES (?, ?, ?, ?)
""", [
    (1, "Alex Rivera", "Los Angeles", 25),
    (2, "Maria Chen", "Santa Monica", 31),
    (3, "Daniel Smith", "Los Angeles", 42),
    (4, "Sarah Johnson", "Pasadena", 28),
    (5, "Anthony Brown", "Long Beach", 35),
    (6, "Jennifer Davis", "Los Angeles", 2
    (7, "Andrew Wilson", "Santa Monica", 3
    (8, "Emily Garcia", "Pasadena", 27)
])
```

```
In [ ]: pd.read_sql("SELECT * FROM customers", con
```

```
In [ ]: pd.read_sql("""
SELECT *
```

```
FROM customers
WHERE city = 'Los Angeles'
""", conn)
```

```
In [ ]: pd.read_sql("""
SELECT *
FROM customers
ORDER BY age ASC
""", conn)
```

```
In [ ]: pd.read_sql("""
SELECT *
FROM customers
WHERE name LIKE '%an%'
""", conn)
```

```
In [ ]: city_counts = pd.read_sql("""
SELECT city, COUNT(*) AS customer_count
FROM customers
GROUP BY city
""", conn)

city_counts
```

```
In [ ]: city_counts.plot(
    kind="bar",
    x="city",
    y="customer_count",
    legend=False
)

plt.title("Number of Customers by City")
plt.xlabel("City")
plt.ylabel("Number of Customers")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
In [ ]: conn.commit()
```

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In [ ]:
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